

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

A piece of wire with a length of **32** inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is **4** more than **3** times the value of y . What is the value of x ?

Correct Answer: 25

Rationale

The correct answer is **25**. It's given that a piece of wire has a length of **32** inches and is cut into two parts. It's also given that one part has a length of x inches and the other part has a length of y inches. It follows that the equation $x + y = 32$ represents this situation. It's also given that the value of x is **4** more than **3** times the value of y , or $x = 3y + 4$. Substituting $3y + 4$ for x in the equation $x + y = 32$ yields $3y + 4 + y = 32$. Combining like terms on the left-hand side of this equation yields $4y + 4 = 32$. Subtracting **4** from both sides of this equation yields $4y = 28$. Dividing both sides of this equation by **4** yields $y = 7$. Substituting **7** for y in the equation $x = 3y + 4$ yields $x = 3(7) + 4$, or $x = 25$. Therefore, the value of x is **25**.